

PRAVIN KUMAR

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OBJECTIVE

Assistant Professor and Ph.D. scholar in medical image analysis, deep learning, and computer vision, with teaching experience across algorithms, data structures, Python programming, pattern recognition, and deep learning. Research interests include diabetic retinopathy screening, lesion segmentation, fundus image analysis, interpretable AI, transformers, generative AI, and reliable AI for healthcare.

RESEARCH INTERESTS

- **Deep Learning, Medical Image Analysis, Computer Vision, Generative AI.**

SKILLS

Facial Recognition, Object Tracking, Video Surveillance, TensorFlow, Pytorch, Python, C++, OpenCV, Pandas, numpy, scikit-learn, OOPs, Data Structures and Algorithms, DBMS.

EXPERIENCE

Sri Balaji University, Pune

Assistant Professor

Pune, Maharashtra

Nov 2024 - Present

- **University SPOC for Smart India Hackathon (SIH):** Organized the event for the entire university and acted as the main contact between our institution and the Ministry of Education.
- **Head, Techno Tuner Club (2025-2026):** Leading technical initiatives and student innovation.
- Deliver lectures on **C Language, Python, Data Structures, Design and Analysis of Algorithms, Theory of Computation, Compiler Design, Pattern Recognition using Deep Learning, and Computer Vision.**
- Develop original course materials and evaluate performance for undergraduate/postgraduate students.

Galgotias University

Assistant Professor

Greater Noida, U.P.

Aug 2023 - Oct 2024

- Conducted engaging lectures on **Data Structures, Algorithms, and Python Programming.**
- Provided academic mentorship and guidance to students.

Ispeck Digital Solutions Private Limited

Data Scientist

Kolkata, West Bengal

March 2023 - June 2023

- Developed deep learning solutions for Face Recognition and YOLO-based PPE compliance.
- Enhanced site security and management through robust computer vision implementation.

EDUCATION

PhD. [Medical Image Analysis], Department of Information Technology

Indian Institute of Information Technology Allahabad, U.P. INDIA.

Pursuing

M.Tech in Human-computer interaction, Department of Information Technology

Indian Institute of Information Technology Allahabad, U.P. INDIA. **[CGPA: 8.00]**

July 2016 - July 2018

MCA B-Level, Information Technology

National Institute of Electronics and Information Technology, New Delhi, INDIA.

Aug 2012 - June 2015

DOEACC A-Level, Information Technology
National Institute of Electronics and Information Technology, New Delhi, INDIA.

Jan 2010 - June 2011

DOEACC O-Level, Information Technology
National Institute of Electronics and Information Technology, New Delhi, INDIA.

Aug 2008 - Dec 2009

Higher Secondary School, Bihar State board
RLSA Mahavidyalaya Vaishali, BIHAR, INDIA.

March 2007

School, Bihar State board
RVMS Maker Saran, BIHAR, INDIA.

March 2005

PUBLICATIONS

1. Pravin Kumar, Satish Kumar Singh, "Cross-Dataset Diabetic Retinopathy Grading with a Lesion-Anchored Multi-View Transformer and Ordinal-Distance Knowledge Distillation", ICT Express (Elsevier), 2026. [Published]

DOI: [10.1016/j.ict.2026.06.010](https://doi.org/10.1016/j.ict.2026.06.010)

- Proposed a five-view, lesion-anchored ConvNeXt-Small transformer that fuses Grad-CAM lesion crops with the full fundus image and distills ordinal CORAL thresholds for robust cross-dataset DR grading.
- Trained once on APTOS, the model reaches 0.92 quadratic weighted kappa in-domain and holds 0.81–0.89 across three unseen datasets.

2. Pravin Kumar, Satish Kumar Singh, "Lesion-Graph Ordinal Reasoning with Calibrated Deferral for Cross-Dataset Diabetic Retinopathy Triage", 11th International Conference on Computer Vision and Image Processing (CVIP 2026), Springer LNCS. [Accepted]

- Introduced a compact post-hoc layer (LGOR-DR) linking calibrated grade probabilities with Core-4 lesion signals to produce a predicted grade, calibrated confidence, and a deferral score for safe triage.
- Calibrated deferral improves retained-case QWK across DeepDRiD, IDRiD, and DDR; deferred cases are enriched for errors by about $1.6\times$.

3. Pravin Kumar, Satish Kumar Singh, "LCK-CBM: Typed Lesion Keypoints for Interpretable Cross-Dataset Diabetic Retinopathy Grading", 11th International Conference on Computer Vision and Image Processing (CVIP 2026), Springer LNCS. [Accepted]

- Proposed a Lesion-Concept-Keypoint concept-bottleneck model placing $K=4$ soft spatial peaks per lesion type, giving 16 clinically interpretable evidence locations and a per-decision audit trail.
- LCK-CBM with self-distillation improves QWK by $+0.031$ over a matched soft-attention CBM on the strict APTOS-2019→IDRiD transfer protocol.

4. Pravin Kumar, Satish Kumar Singh, "Parameter-Efficient Dual-Attention Microaneurysm Segmentation with Robust Cross-Dataset Generalization", Biomedical Signal Processing and Control (Elsevier). [Under Review]

- Designed MSDAG-Net, a compact 5.3M-parameter dual-attention encoder-decoder with an SE-weighted atrous-pyramid bottleneck, skip gates, and deep supervision for microaneurysm segmentation.
- Under a strict official-test protocol, delivers the best cross-dataset generalization (IDRiD→FGADR AUC-PR 0.58–0.60) at a fraction of transformer-model parameters.

ACADEMIC ACCOMPLISHMENTS TRAINING

- Attended workshop on Advances in Deep Architectures for Signal, Image and Vision Applications (ADASIVA - 2018).
- Associated serving as teaching assistant for Deep-learning, Computer graphics, Image and video processing, Data compression, C (Aug 2019 - March 2023) language programming.

- Being mentor for interns and M-tech student in Computer Vision Biometric Lab (CVBL)(Aug 2019 - March 2023).
- Secured All India Rank 4619 among 100,000 students appeared for GATE-2016.

EXTRA-CURRICULAR ACTIVITIES

- Assisted in organizing and hospitality Coordinator for the IEEE Signal Processing Society meet and Summer School on Advances in Deep Architectures for Signal, Image and Vision Applications(ADASIVA) - 2018, 2019, 2020, 2021 and 2022.
- Assisted in Organizing Computer Vision and Image Processing - 5th International Conference in IIIT Allahabad 2021.
- IEEE Treasurer UP Section IIIT Allahabad, from 2019 - 2022.
- Presidential council member of Gymkhana and PhD. representative, from 2020 - 2022.
- **Digital Education Outreach** (in progress). [CSwithPravin](#) (March 2020 - present)